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1. Introduction

This document models connected coffee machines that send `telemetry` to a `central cloud site`, receive OTA firmware updates, and produce operational and audit logging evidence. Authored functional architecture remains stable while runtime, code, and verification layers are inferred from infrastructure, source, and test artifacts.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	stable	Stable authored capability boundaries and responsibility ownership.	7	7	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (`VIEW-COMMUNICATION`)	in-review	Communication handoffs are mapped; protocol detail remains partially inferred.	7	5	2	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (`VIEW-DEPLOYMENT`)	in-review	Deployment/runtime mapping is present with evolving control-plane evidence depth.	7	5	2	medium	Counts a unit as covered when direct inferred runtime evidence exists.

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Security View (`VIEW-SECURITY`)	draft	Core threats are modeled while additional control evidence is being expanded.	7	7	1	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Requirement mappings exist; verification and evidence coverage continue to be hardened.	7	7	2	high	Counts a unit as covered when direct inferred runtime or code evidence exists.

4. Terms and Definitions

Term	Definition
`actor` (EM-ACTOR)	A person or operational role that interacts with functional units.
`attack vector` (EM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
`audit log record` (DATA-AUDIT-LOG-RECORD)	Immutable log record for security and compliance analysis. Aliases:audit records, logging records
`audit logging is enabled` (FEAT-AUDIT-LOGGING)	Feature flag enabling audit-grade log persistence. Aliases:audit enabled
`authored mapping` (EM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
`barista` (ACT-BARISTA)	Operates coffee machines and observes local machine health. Aliases:store operator
`brew cycle completed event is received` (EVT-BREW-CYCLE-COMPLETED)	Event raised when a coffee machine completes a brew cycle. Aliases:brew completed, brew cycle event
`brew telemetry` (DATA-BREW-TELEMETRY)	Machine telemetry including brew duration, temperature, and status metadata. Aliases:machine status, machine telemetry, telemetry, telemetry payloads, telemetry records
`campaign id` (DATA-CAMPAIGN-ID)	Stable OTA campaign identifier used to correlate rollout and audit records. Aliases:campaign identifier, campaign ids
`cloud control` (FG-CLOUD-CONTROL)	Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Aliases:central cloud site
`cloud ingest endpoint is unavailable` (EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE)	Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Aliases:endpoint outage, ingest endpoint unavailable

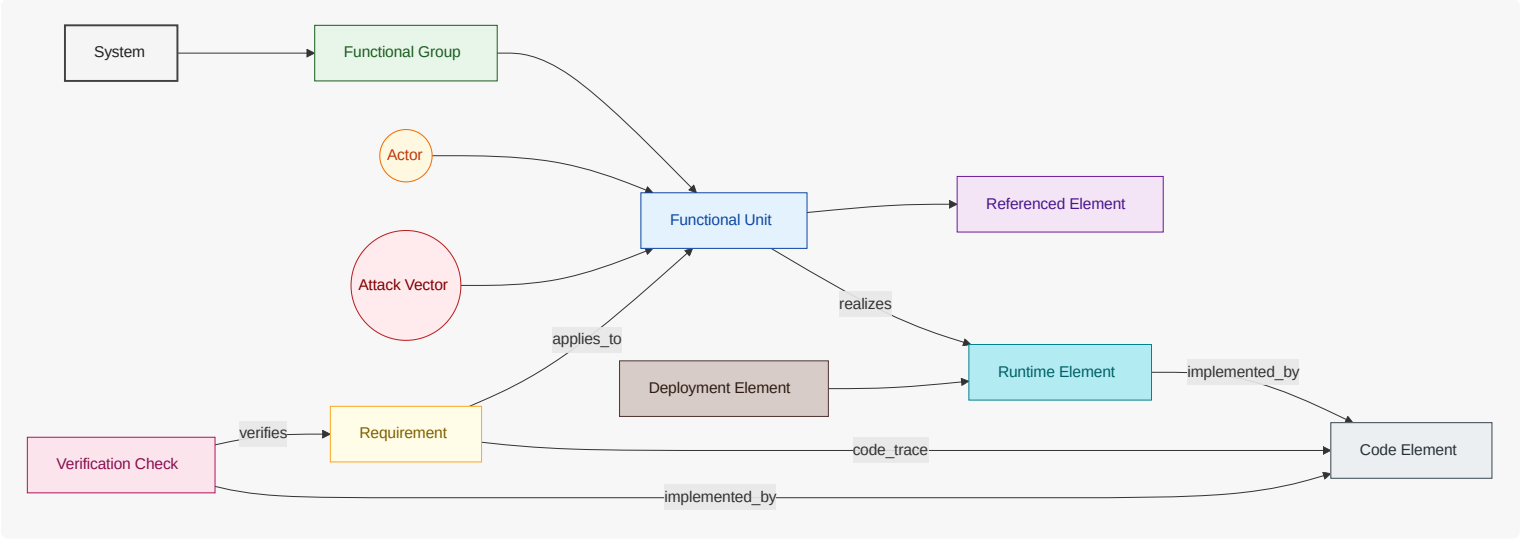
Term	Definition
`cloud logging service` (REF-CLOUD-LOGGING-SERVICE)	Managed log and metrics service used for fleet observability. Aliases:logging service
`cloud runtime operations` (FU-CLOUD-RUNTIME-OPERATIONS)	Operates cloud runtime deployment and scaling controls. Aliases:runtime operations
`cloud runtime sdk` (REF-CLOUD-RUNTIME-SDK)	SDK used by cloud units for runtime integrations. Aliases:cloud sdk
`code element` (EM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
`coffee fleet system` (SYS-COFFEE-FLEET-SYSTEM)	Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Aliases:the coffee fleet system
`coffee machine fleet operation is active` (STATE-FLEET-OPERATION-ACTIVE)	State where machines and central cloud are actively serving operations. Aliases:fleet operation active
`device identity and secrets` (FU-DEVICE-IDENTITY-SECRETS)	Manages device identity material and secure configuration. Aliases:device secrets
`firmware bundle store` (REF-FIRMWARE-BUNDLE-STORE)	Object store containing signed OTA firmware bundles. Aliases:firmware store
`firmware signature is valid` (COND-FIRMWARE-SIGNATURE-VALID)	Condition indicating firmware signature verification succeeded. Aliases:signature valid
`firmware signer service` (REF-FIRMWARE-SIGNER-SERVICE)	Service that signs firmware artifacts before OTA rollout. Aliases:signer service
`firmware validation failed event is received` (EVT-FIRMWARE-VALIDATION-FAILED)	Event raised when signature/hash validation fails on firmware. Aliases:validation failed
`fleet ingestion api` (FU-FLEET-INGESTION-API)	Ingests telemetry and machine status from edge devices. Aliases:ingestion api
`fleet observability reporting` (FU-FLEET-OBSERVABILITY-REPORTING)	Aggregates telemetry and update outcomes into operational and audit signals. Aliases:observability reporting
`fleet operator` (ACT-FLEET-OPERATOR)	Owns OTA rollout campaigns and fleet-level reliability. Aliases:fleet admin
`functional group` (EM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
`functional unit` (EM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
`inferred mapping` (EM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.

Term	Definition
`iot ingest endpoint` (REF-IOT-INGEST-ENDPOINT)	Cloud endpoint for receiving telemetry and status from devices. Aliases:iot endpoint
`log tampering` (AV-LOG-TAMPERING)	Attempts to suppress or alter security/operational log records. Aliases:audit tampering
`machine edge` (FG-MACHINE-EDGE)	Capability area for device-local telemetry and OTA execution behavior. Aliases:coffee machine edge
`machine identity` (DATA-MACHINE-IDENTITY)	Device identity material used for authenticated publish, ingest, and audit attribution. Aliases:device identity, machine identities
`machine telemetry collection` (FU-MACHINE-TELEMETRY-COLLECTION)	Collects brew/device telemetry and publishes cloud-bound payloads. Aliases:telemetry collection
`malicious firmware bundle` (AV-MALICIOUS-FIRMWARE-BUNDLE)	Tampered firmware payload attempts to execute on devices. Aliases:firmware tampering
`mqtt device sdk` (REF-MQTT-DEVICE-SDK)	SDK used by edge components for telemetry and command transport. Aliases:mqtt sdk
`normal mode is active` (MODE-NORMAL)	Default operation mode for telemetry and OTA behavior. Aliases:normal mode, normal operation mode
`ota campaign is enabled` (FEAT-OTA-CAMPAIGN)	Feature flag enabling centralized OTA rollout behavior. Aliases:ota enabled, ota rollout
`ota campaign policy` (DATA-OTA-CAMPAIGN-POLICY)	Policy describing rollout cohorts, windows, and rollback criteria. Aliases:campaign policy, rollout cohorts, rollout windows
`ota campaign scheduled event is received` (EVT-OTA-CAMPAIGN-SCHEDULED)	Event raised when a fleet OTA campaign is scheduled. Aliases:campaign commands, campaign scheduled, ota commands, ota rollout commands
`ota status` (DATA-OTA-STATUS)	Device-level status for firmware download, apply, and rollback outcomes. Aliases:firmware apply, ota outcomes, rollback firmware apply, update outcomes, update/rollback status
`ota status reported event is received` (EVT-OTA-STATUS-REPORTED)	Event raised when device reports OTA apply status. Aliases:apply status, apply/rollback status, ota status reported, rollback status, status feedback, update status
`ota update agent` (FU-OTA-UPDATE-AGENT)	Downloads, validates, and applies signed firmware updates on machines. Aliases:device update agent
`platform operations` (FG-PLATFORM-OPERATIONS)	Capability area for runtime operations and device identity/secrets lifecycle. Aliases:fleet platform operations
`platform operator` (ACT-PLATFORM-OPERATOR)	Owns runtime deployment and operational controls. Aliases:platform ops
`referenced element` (EM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.

Term	Definition
`requirement` (EM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
`runtime element` (EM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
`security analyst` (ACT-SECURITY-ANALYST)	Reviews identity, firmware integrity, and logging controls. Aliases:security reviewer
`signed firmware bundle` (DATA-FIRMWARE-BUNDLE)	Signed OTA firmware artifact distributed to machine cohorts for update apply. Aliases:firmware artifact, firmware bundle, firmware bundles, signed firmware bundles
`spoofed device identity` (AV-SPOOFED-DEVICE-IDENTITY)	Unauthorized actor impersonates a machine identity. Aliases:device spoofing
`telemetry ingested event is received` (EVT-TELEMETRY-INGESTED)	Event raised when cloud ingestion accepts telemetry. Aliases:fleet events, ingestion events, observability event fanout, observability events, telemetry ingested
`telemetry replay abuse` (AV-TELEMETRY-REPLAY-ABUSE)	Replayed telemetry floods ingestion and distorts reporting. Aliases:telemetry replay
`telemetry replay abuse is detected` (EVT-TELEMETRY-REPLAY-ABUSE-DETECTED)	Event raised when replayed telemetry behavior is detected and flagged as abuse. Aliases:replay abuse, telemetry replay detected
`update campaign orchestration` (FU-UPDATE-CAMPAIGN-ORCHESTRATION)	Selects target machine groups and coordinates OTA rollout phases. Aliases:campaign orchestration
`upward linking` (EM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
`verification check` (EM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.

4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



5. Architecture Intent View

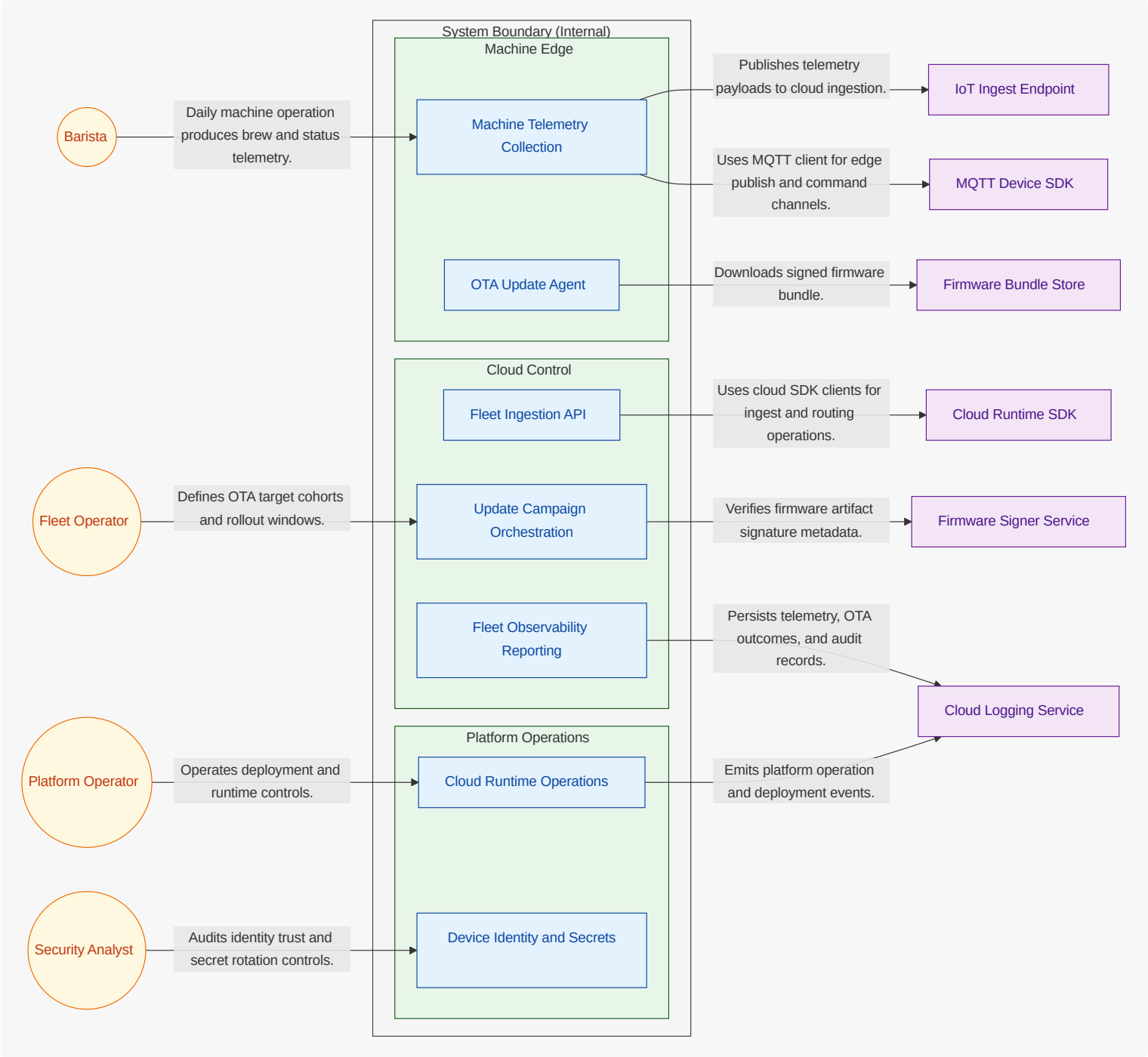
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

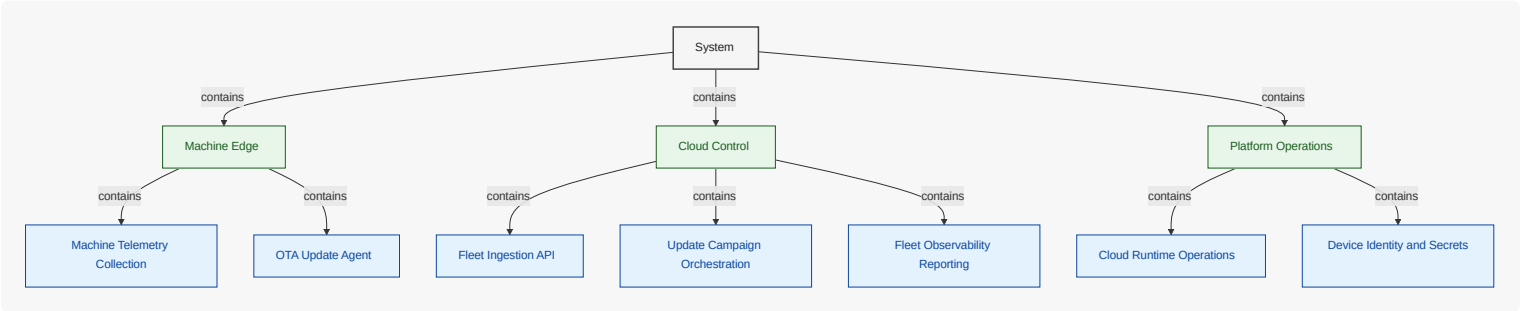
5.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



5.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



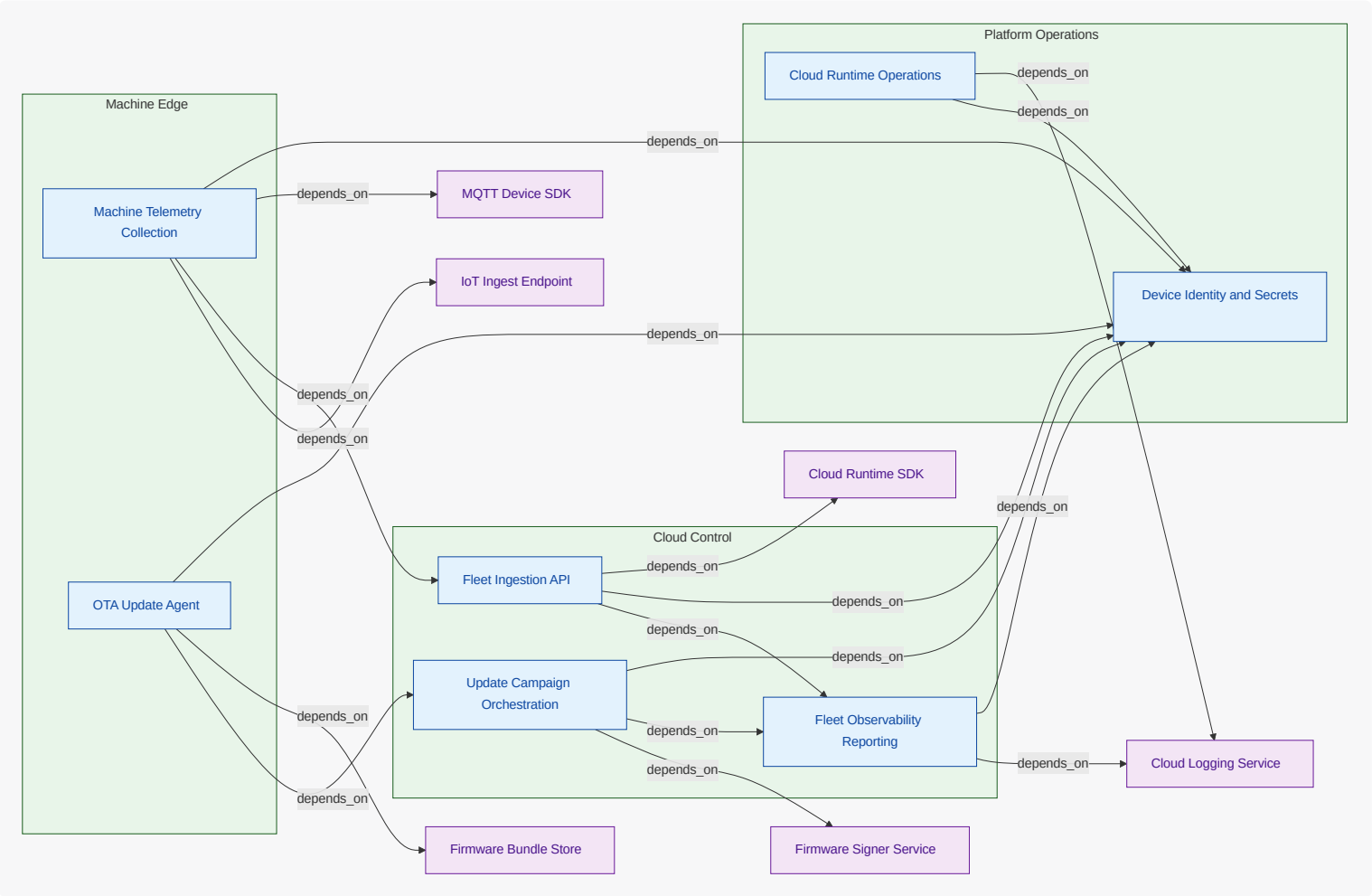
5.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

	Fleet Ingestion API	
Machine Telemetry Collection	Fleet Observability Reporting	Cloud Runtime Operations
OTA Update Agent	Update Campaign Orchestration	Device Identity and Secrets
Machine Edge	Cloud Control	Platform Operations

5.5. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.



5.6. Functional Groups and Units (View Scoped)

5.6.1. Machine Edge (FG-MACHINE-EDGE)

Device-local `telemetry` and OTA execution responsibilities.

`Machine Edge` owns behavior that executes directly on coffee machines. It gathers `telemetry`, applies OTA updates, and reports local status to cloud boundaries.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Collects brew/device `telemetry` and transmits cloud-bound `telemetry payloads`.

Owned Responsibility

functional responsibility for `machine telemetry collection`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Fleet Ingestion API`

Key Collaborators

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Linked Requirements

`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

Downloads `signed firmware bundles`, validates integrity, applies updates, and reports status.

Owned Responsibility

functional responsibility for `ota update agent`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Firmware Bundle Store`

Key Collaborators

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

Linked Requirements

`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`

5.6.2. Cloud Control (FG-CLOUD-CONTROL)

Central cloud ingestion, campaign control, and fleet reporting.

`Cloud Control` owns central APIs and coordination logic for fleet `telemetry` and `OTA rollouts`. It provides global fleet visibility and deterministic rollout decisions.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Receives `telemetry` and `machine status`, applies validation, and emits `ingestion events`.

Owned Responsibility

functional responsibility for `fleet ingestion api`; includes decision and orchestration flow to `Cloud Runtime SDK` and decision and orchestration flow to `Device Identity and Secrets`

Key Collaborators

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-011`, `REQ-COF-012`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Aggregates `telemetry` and `OTA outcomes` into dashboards, alerts, and `audit records`.

Owned Responsibility

functional responsibility for `fleet observability reporting`; includes decision and orchestration flow to `Cloud Logging Service` and decision and orchestration flow to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Plans OTA `rollout cohorts`, drives update commands, and handles rollback flow decisions.

Owned Responsibility

functional responsibility for `update campaign orchestration`; includes decision and orchestration flow to `Device Identity and Secrets` and decision and orchestration flow to `Firmware Signer Service`

Key Collaborators

`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-003`, `REQ-COF-005`, `REQ-COF-010`

5.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Cloud runtime operations` plus identity and secret-management controls.

`Platform Operations` owns deployability, runtime guardrails, and secure identity/config lifecycle. It ensures cloud and device trust boundaries remain enforceable over time.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

Owned Responsibility

functional responsibility for `cloud runtime operations`; includes decision and orchestration flow to `Cloud Logging Service` and decision and orchestration flow to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-008`, `REQ-COF-012`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

Owned Responsibility

functional responsibility for `device identity and secrets`

Key Collaborators

none

Linked Requirements

`REQ-COF-004`, `REQ-COF-007`

6. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

6.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.

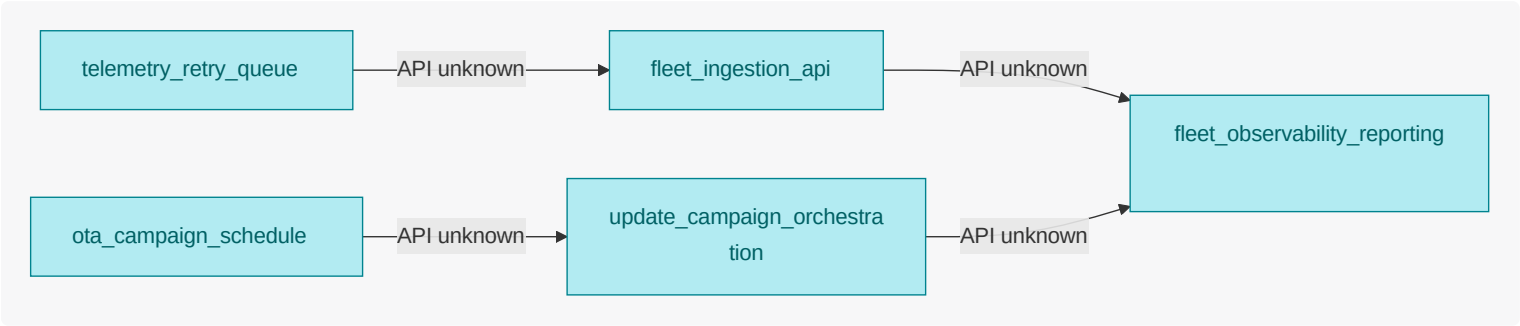
6.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/event/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

6.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE**
> This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



6.5. Functional Groups and Units (View Scoped)

6.5.1. Machine Edge (FG-MACHINE-EDGE)

`Machine Edge` sends `telemetry` upstream, receives `OTA commands`, and reports `apply/rollback status`.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Sends `telemetry` over authenticated channels and retries during `endpoint outage` windows.

Inbound Interfaces

Interface	Description
`Barista`	Daily machine operation produces brew and status `telemetry`.

Outbound Interfaces

Interface	Description
`Device Identity and Secrets`	Uses `machine identity` material for authenticated publish.
`Fleet Ingestion API`	Sends normalized `telemetry records` to `ingestion API`.
`IoT Ingest Endpoint`	Publishes `telemetry payloads` to cloud ingestion.
`MQTT Device SDK`	Uses MQTT client for edge publish and command channels.

Communication Evidence

runtime: `telemetry_retry_queue` | code modules: `machine_telemetry`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

Receives `campaign commands` and reports `update/rollback status` back to `cloud control`.

Inbound Interfaces

none

Outbound Interfaces

Interface	Description
`Device Identity and Secrets`	Uses trust anchors for firmware and endpoint authentication.
`Firmware Bundle Store`	Downloads `signed firmware bundle`.
`Update Campaign Orchestration`	Receives `OTA rollout commands` and policy constraints.

Communication Evidence

runtime: `ota_campaign_schedule`, `ota_campaign_target` | code modules: `ota_update_agent`

6.5.2. Cloud Control (FG-CLOUD-CONTROL)

`Cloud Control` coordinates `telemetry` intake, `campaign commands`, and `observability event fanout`.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Accepts `machine telemetry` and forwards normalized records to `observability reporting`.

Inbound Interfaces

Interface	Description
`Machine Telemetry Collection`	Sends normalized `telemetry records` to `ingestion API`.

Outbound Interfaces

Interface	Description
`Cloud Runtime SDK`	Uses `cloud SDK` clients for ingest and routing operations.
`Device Identity and Secrets`	Validates `device identity` and token claims.
`Fleet Observability Reporting`	Forwards validated `telemetry` and `machine status` events.

Communication Evidence

runtime: `fleet_event_topic`, `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Consumes `telemetry` and `OTA outcomes` and emits dashboards/alerts/log events.

Inbound Interfaces

Interface	Description
`Fleet Ingestion API`	Forwards validated `telemetry` and `machine status` events.
`Update Campaign Orchestration`	Emits `OTA rollout` and rollback events for fleet visibility.

Outbound Interfaces

Interface	Description
`Cloud Logging Service`	Persists `telemetry`, `OTA outcomes`, and `audit records`.
`Device Identity and Secrets`	Uses controlled credentials for audit-log writes.

Communication Evidence

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Sends `OTA commands` to update agents and ingests `status feedback` for campaign control.

Inbound Interfaces

Interface	Description
`Fleet Operator`	Defines OTA target cohorts and `rollout windows`.
`OTA Update Agent`	Receives `OTA rollout commands` and policy constraints.

Outbound Interfaces

Interface	Description
`Device Identity and Secrets`	Uses trusted control-plane credentials.
`Firmware Signer Service`	Verifies `firmware artifact` signature metadata.
`Fleet Observability Reporting`	Emits `OTA rollout` and rollback events for fleet visibility.

Communication Evidence

runtime: `update_campaign_orchestration` | code modules: `update_campaign_orchestrator`

6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform operations` coordinates runtime controls and secret distribution contracts with dependent units.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Coordinates deployment/runtime controls that shape fleet service availability.

Inbound Interfaces

Interface	Description
`Platform Operator`	Operates deployment and runtime controls.

Outbound Interfaces

Interface	Description
`Cloud Logging Service`	Emits platform operation and deployment events.
`Device Identity and Secrets`	Requires runtime credentials and encrypted config.

Communication Evidence

code modules: `cloud_runtime_ops`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Provides identity validation and secure config interfaces to edge and cloud units.

Inbound Interfaces

Interface	Description
`Cloud Runtime Operations`	Requires runtime credentials and encrypted config.
`Fleet Ingestion API`	Validates `device identity` and token claims.
`Fleet Observability Reporting`	Uses controlled credentials for audit-log writes.
`Machine Telemetry Collection`	Uses `machine identity` material for authenticated publish.
`OTA Update Agent`	Uses trust anchors for firmware and endpoint authentication.
`Security Analyst`	Audits identity trust and secret rotation controls.
`Update Campaign Orchestration`	Uses trusted control-plane credentials.

Outbound Interfaces

none

Communication Evidence

code modules: `device_identity_secrets`

7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and `platform operations`.

7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

7.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.

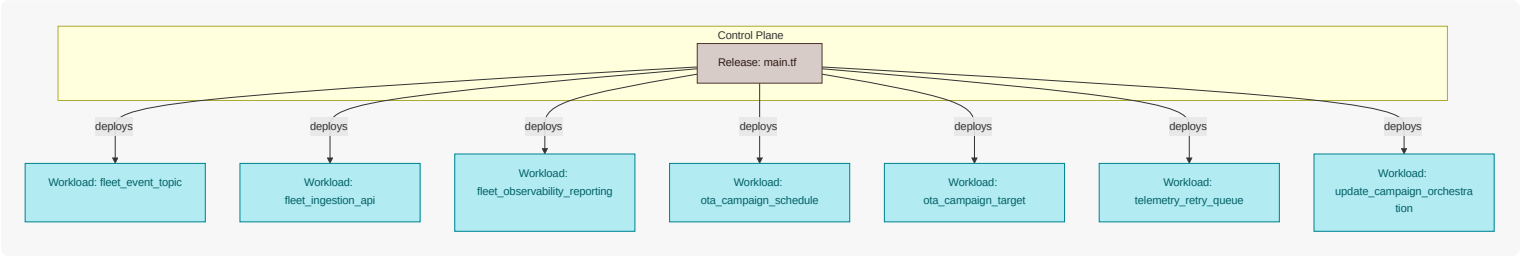
7.3. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

7.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

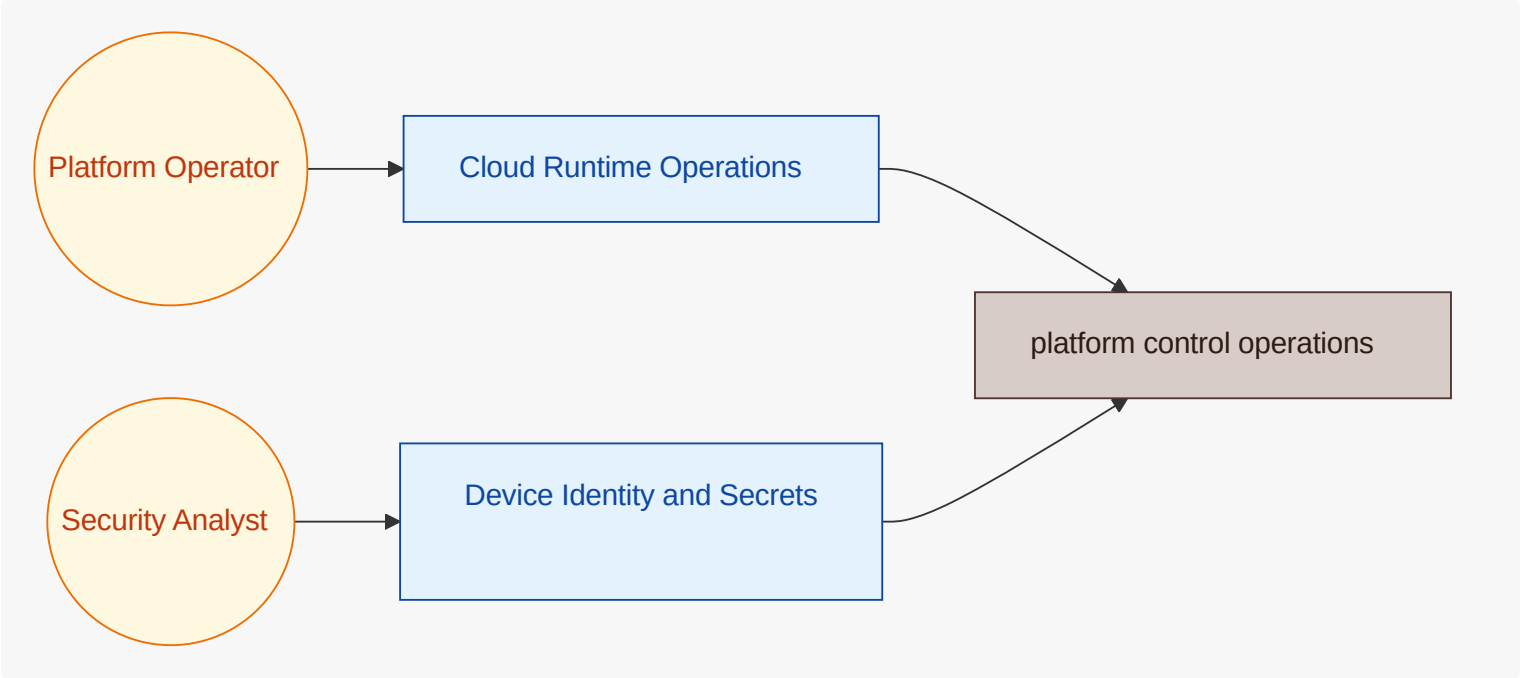
> **NOTE**
> This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.



7.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

> **NOTE**
> Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



7.6. Functional Groups and Units (View Scoped)

7.6.1. Machine Edge (FG-MACHINE-EDGE)

Machine-edge software artifacts are rolled independently from `cloud control` to isolate device risk.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Machine Telemetry Collection Deployment

Deployed as edge software component with independent release cadence.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`
Deployment dependencies	`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`
Inferred deployment evidence	runtime: `telemetry_retry_queue`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

OTA Update Agent Deployment

Versioned and rolled independently from `telemetry` pipeline components.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`
Deployment dependencies	`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`
Inferred deployment evidence	runtime: `ota_campaign_schedule`, `ota_campaign_target`

7.6.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud-control units deploy independently to reduce blast radius during campaign logic changes.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Fleet Ingestion API Deployment

Deployed as independent cloud ingestion boundary to isolate scaling concerns.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-011`, `REQ-COF-012`
Deployment dependencies	`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`
Inferred deployment evidence	runtime: `fleet_event_topic`, `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Fleet Observability Reporting Deployment

Deployed as reporting plane with independent scaling and retention controls.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`
Deployment dependencies	`Cloud Logging Service`, `Device Identity and Secrets`
Inferred deployment evidence	runtime: `fleet_observability_reporting`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Update Campaign Orchestration Deployment

Isolated deployment supports safe OTA policy evolution.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-003`, `REQ-COF-005`, `REQ-COF-010`
Deployment dependencies	`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`
Inferred deployment evidence	runtime: `update_campaign_orchestration`

7.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform operations` governs release sequencing, runtime placement, and operational guardrails.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Cloud Runtime Operations Deployment

Owns source-to-runtime release controls and rollback guardrails.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-008`, `REQ-COF-012`

Deployment Aspect	View-Scoped Detail
Deployment dependencies	`Cloud Logging Service`, `Device Identity and Secrets`
Inferred deployment evidence	code modules: `cloud_runtime_ops`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Device Identity and Secrets Deployment

Managed as shared control-plane boundary with controlled key rotation.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-004`, `REQ-COF-007`
Deployment dependencies	none
Inferred deployment evidence	code modules: `device_identity_secrets`

8. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

8.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

8.2. Coverage Gaps

Coverage summary: 6/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

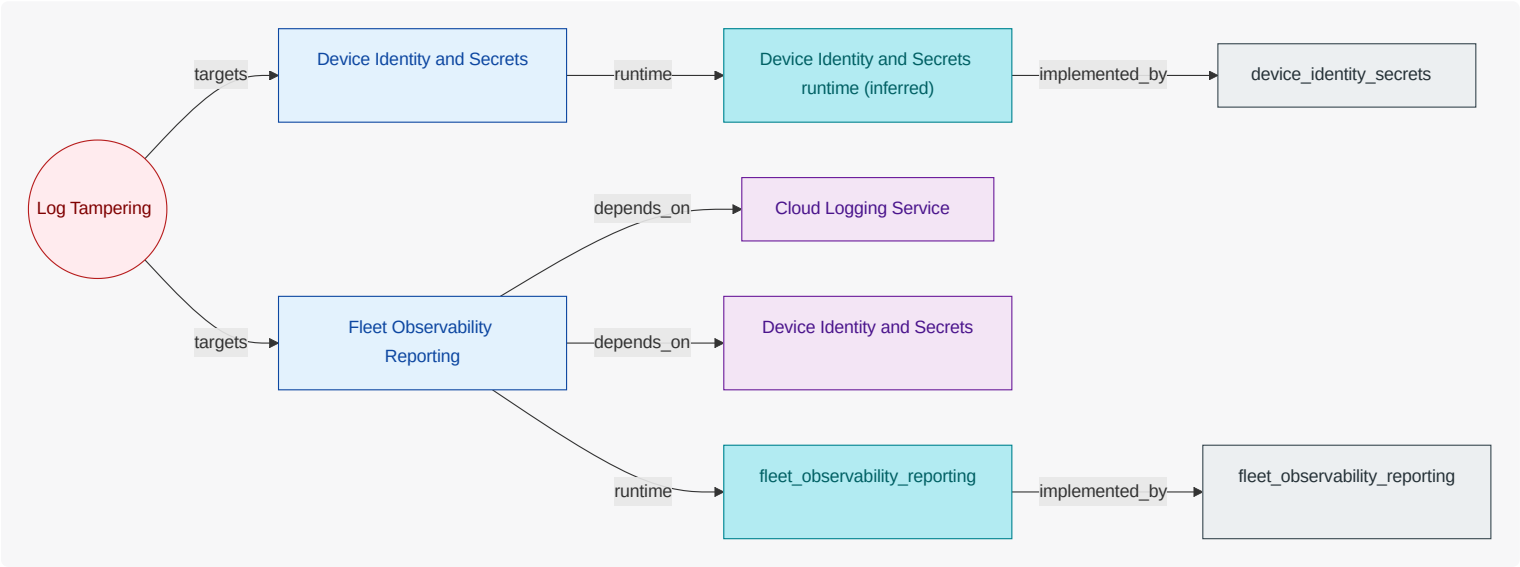
8.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/events) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

8.4. Attack Vector Chapters

8.4.1. `Log Tampering` (AV-LOG-TAMPERING)

Attempts to suppress or alter audit/operational logs.



Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Protects credential lifecycle and enforces least-privilege access patterns.

Attack Surface

none

Security Evidence Present

code modules: `device_identity_secrets`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

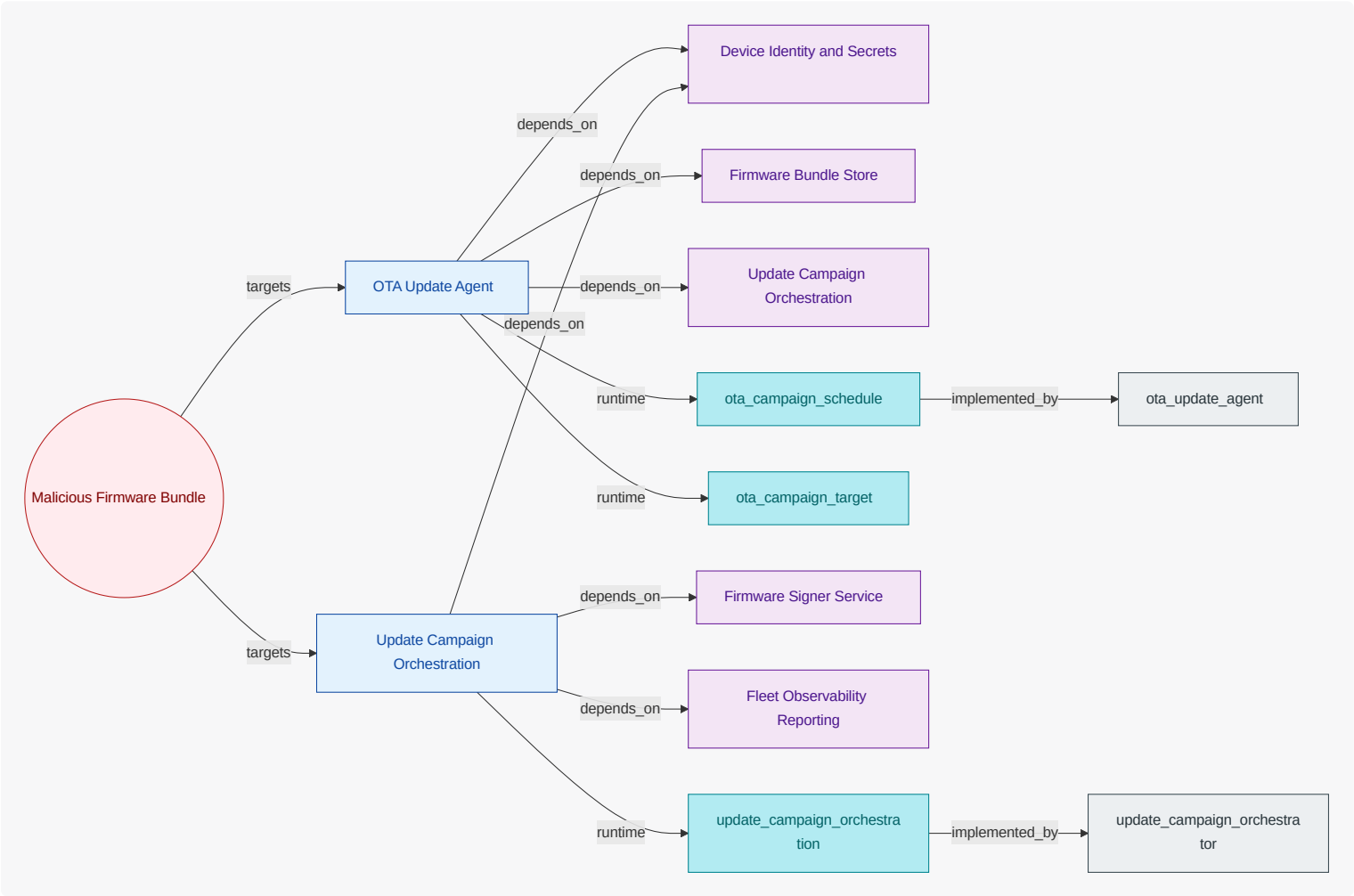
`Cloud Logging Service`, `Device Identity and Secrets`

Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

8.4.2. Malicious Firmware Bundle (AV-MALICIOUS-FIRMWARE-BUNDLE)

Tampered OTA payload targeting update agent execution.



OTA Update Agent (FU-OTA-UPDATE-AGENT)

Enforces `signature validation` and rejects untrusted `firmware bundles`.

Attack Surface

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

Security Evidence Present

runtime: `ota_campaign_schedule`, `ota_campaign_target` | code modules: `ota_update_agent`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Enforces signed-firmware policy and campaign authorization boundaries.

Attack Surface

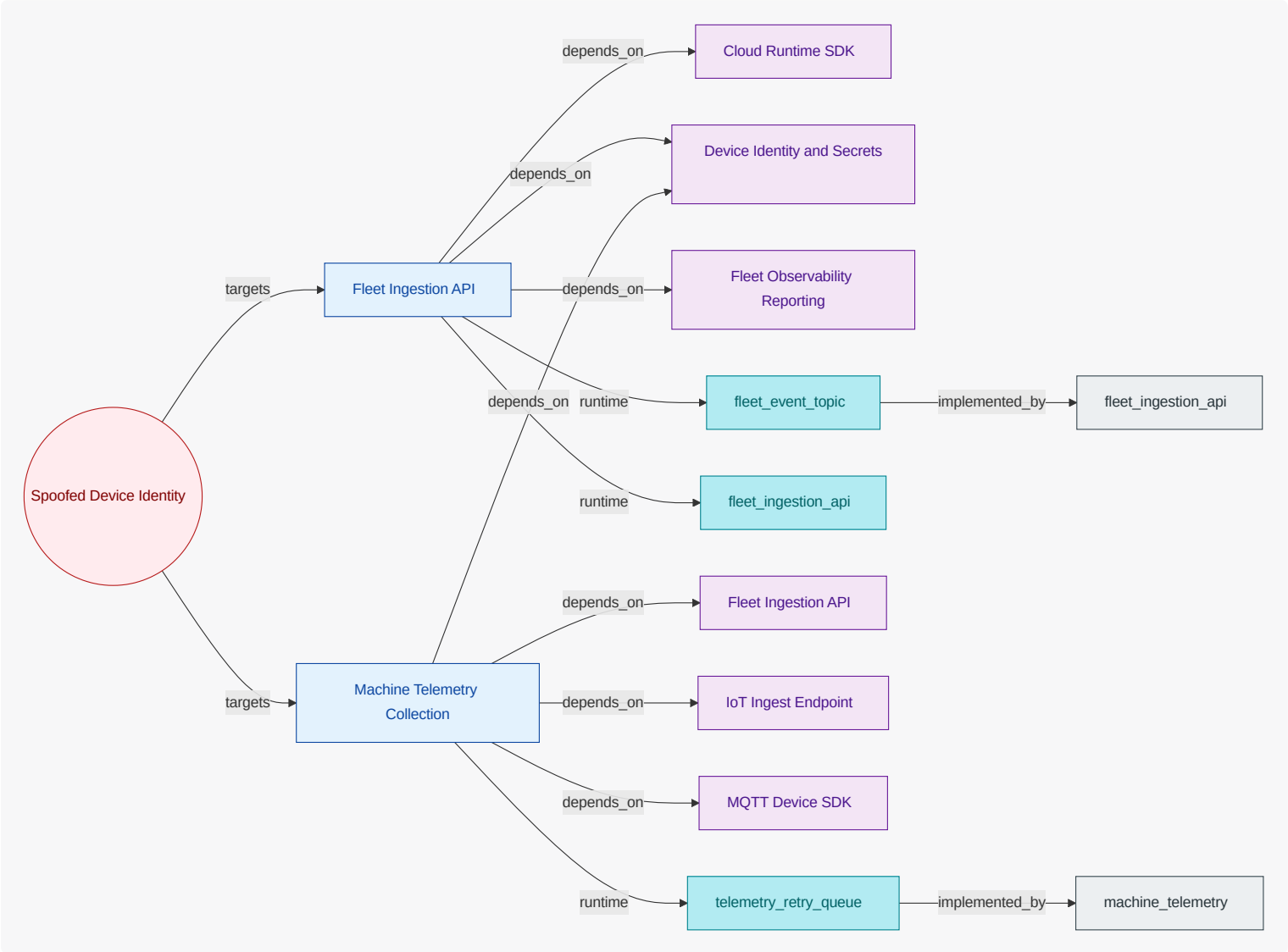
`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `update_campaign_orchestration` | code modules: `update_campaign_orchestrator`

8.4.3. `Spoofed Device Identity` (AV-SPOOFED-DEVICE-IDENTITY)

Unauthorized identity attempting `telemetry`/command access.



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `fleet_event_topic`, `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Protects `telemetry` authenticity and defends against `replay abuse` patterns.

Attack Surface

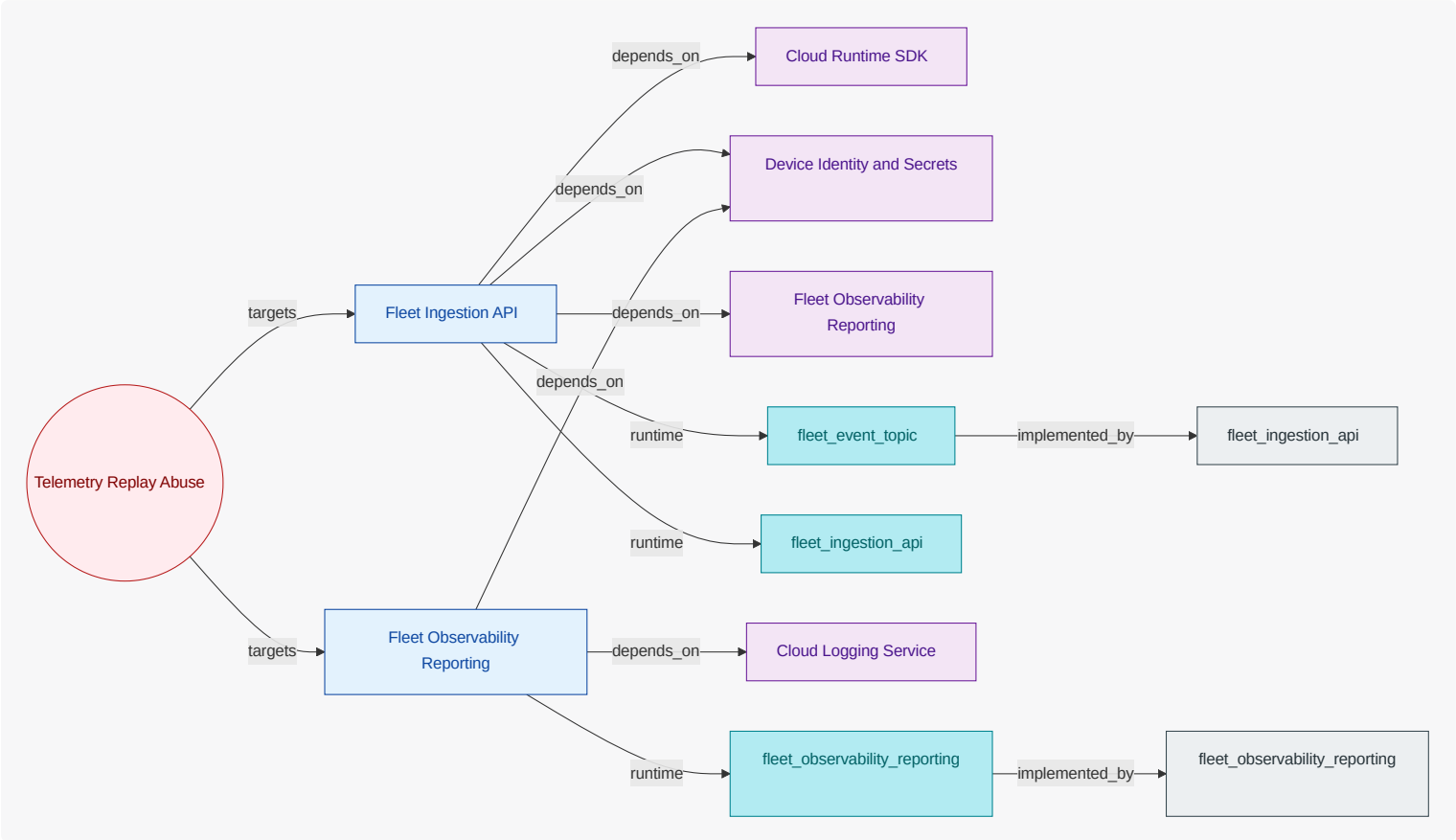
`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Security Evidence Present

runtime: `telemetry_retry_queue` | code modules: `machine_telemetry`

8.4.4. `Telemetry Replay Abuse` (AV-TELEMETRY-REPLAY-ABUSE)

Replayed `telemetry` intended to overload ingestion and corrupt insights.



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `fleet_event_topic`, `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

`Cloud Logging Service`, `Device Identity and Secrets`

Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

9.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

9.2. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- `Cloud Runtime Operations` has no direct inferred runtime evidence yet.
- `Device Identity and Secrets` has no direct inferred runtime evidence yet.

9.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

9.4. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Requirement	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING	FU-MACHINE-TELEMETRY-COLLECTION	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
REQ-COF-001			X		X		
REQ-COF-002			X	X			
REQ-COF-003						X	X
REQ-COF-004		X				X	
REQ-COF-005				X		X	X
REQ-COF-006			X		X		
REQ-COF-007		X		X			
REQ-COF-008	X		X	X			
REQ-COF-009			X	X			
REQ-COF-010				X		X	X

Requirement	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING	FU-MACHINE-TELEMETRY-COLLECTION	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
REQ-COF-011			X		X		
REQ-COF-012	X		X	X			

9.5. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`	`REQ-COF-001`	pass	test-results/contract-telemetry.json ingest payload accepted with `machine identity` and `telemetry` fields
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-002`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-003`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-003`	pass	test-results/unit-ota.xml
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-004`	pass	test-results/unit-ota.xml
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-005`	partial	test-results/ota-rollback-report.md
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-007`	pass	test-results/integration-logging.json `audit log record` persisted with redacted `machine identity`

10. Traceability Appendix

10.1. Requirement Details

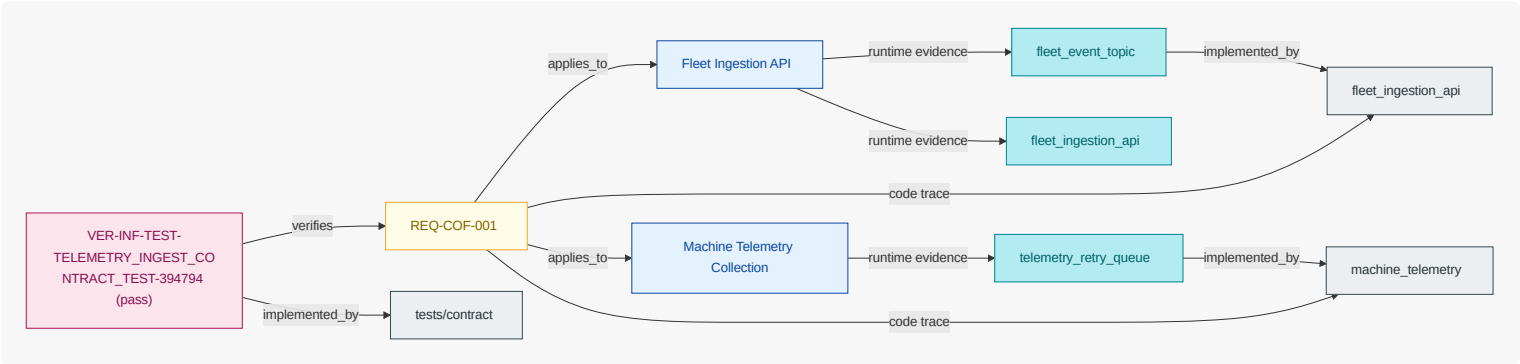
10.1.1. REQ-COF-001

While `coffee machine fleet operation is active`, when `brew cycle completed event is received`, `the coffee fleet system` shall publish `brew telemetry` to `fleet ingestion api`.

> **NOTE** Primary machine-to-cloud `telemetry` contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



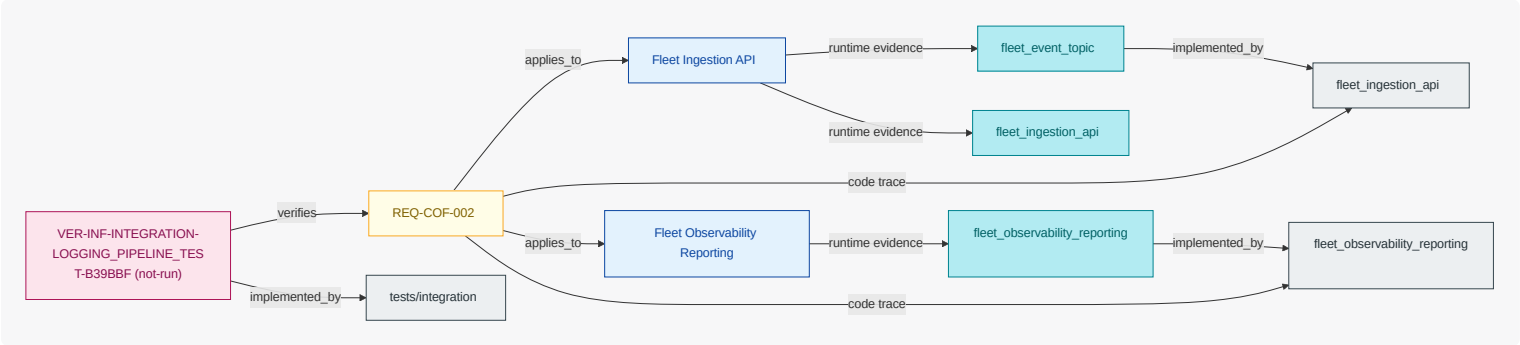
10.1.2. REQ-COF-002

While `coffee machine fleet operation is active`, when `telemetry ingested event is received`, `the coffee fleet system` shall persist fleet `telemetry` metrics for `fleet operator`.

> **NOTE** Operational observability `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



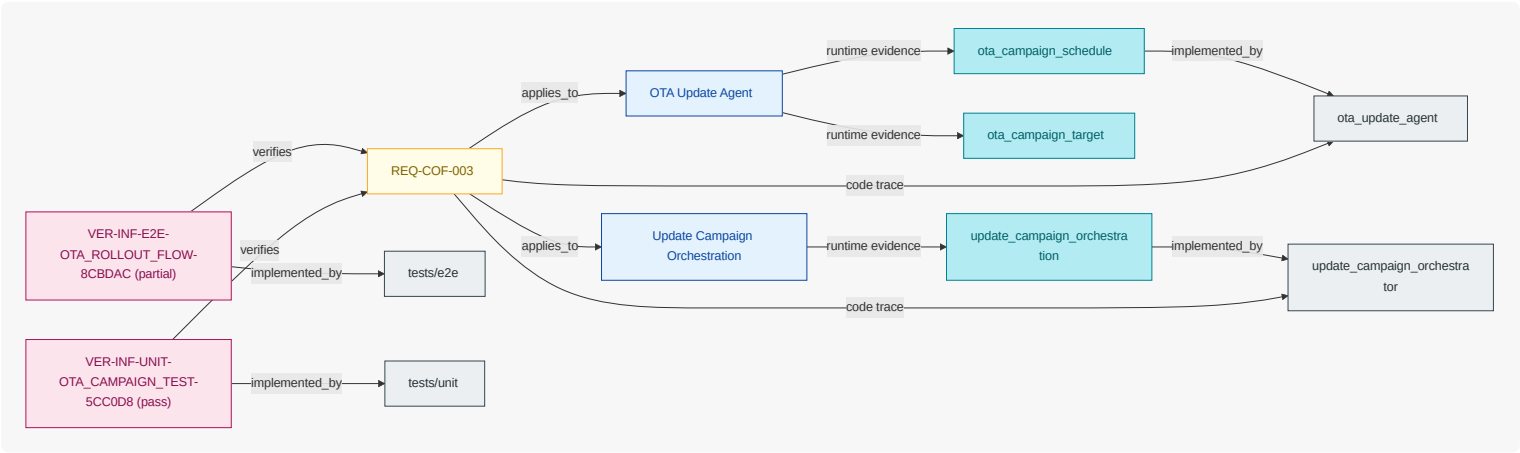
10.1.3. REQ-COF-003

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall distribute `signed firmware bundle` to targeted machine cohort.

> **NOTE** `OTA rollout` behavior from central `cloud control`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



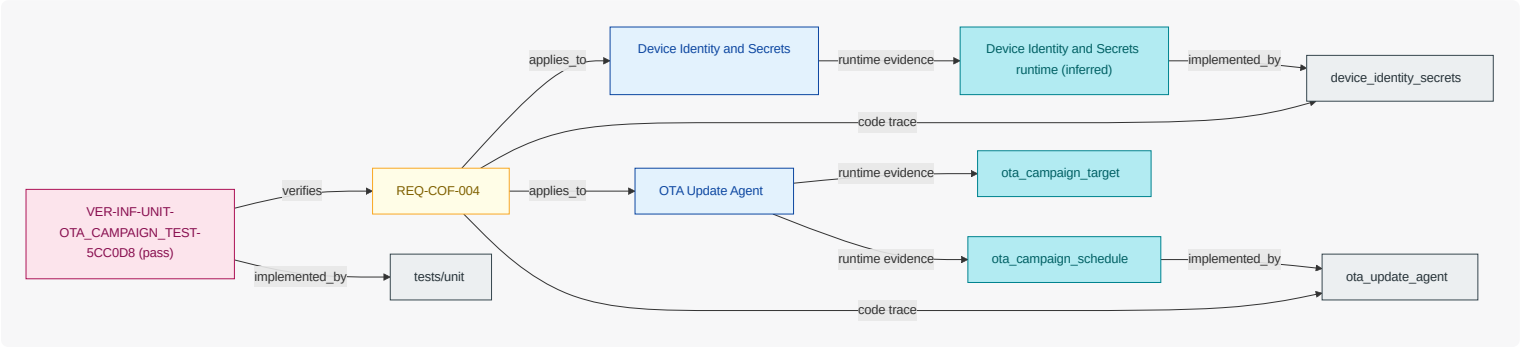
10.1.4. REQ-COF-004

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall gate `firmware apply` on `firmware signature is valid`.

> **NOTE** Firmware integrity and trust boundary control.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



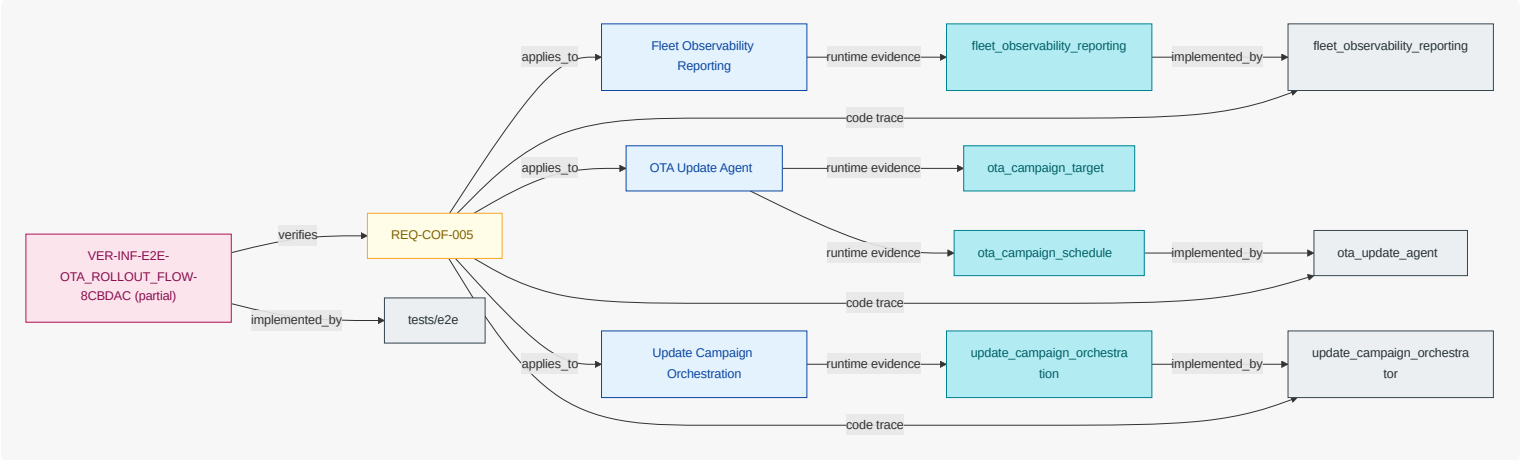
10.1.5. REQ-COF-005

If `firmware validation failed event is received`, then `the coffee fleet system` shall `rollback firmware apply`.

> **NOTE** Rollback and failure signaling path.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



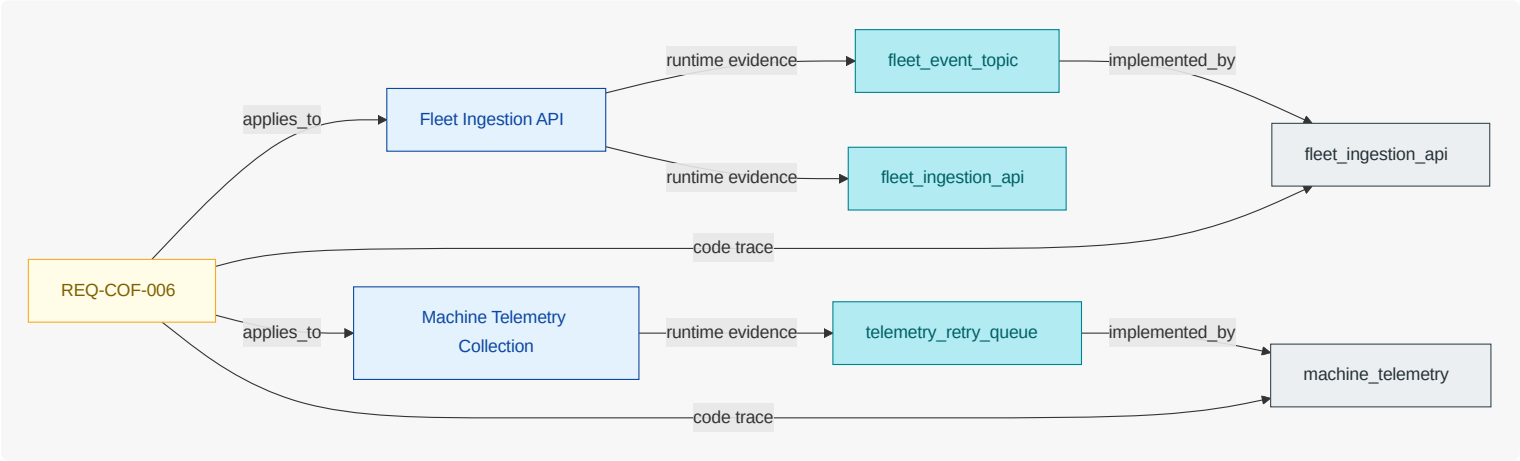
10.1.6. REQ-COF-006

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall queue `telemetry` for retry.

> **NOTE** Resilience for temporary cloud outage.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



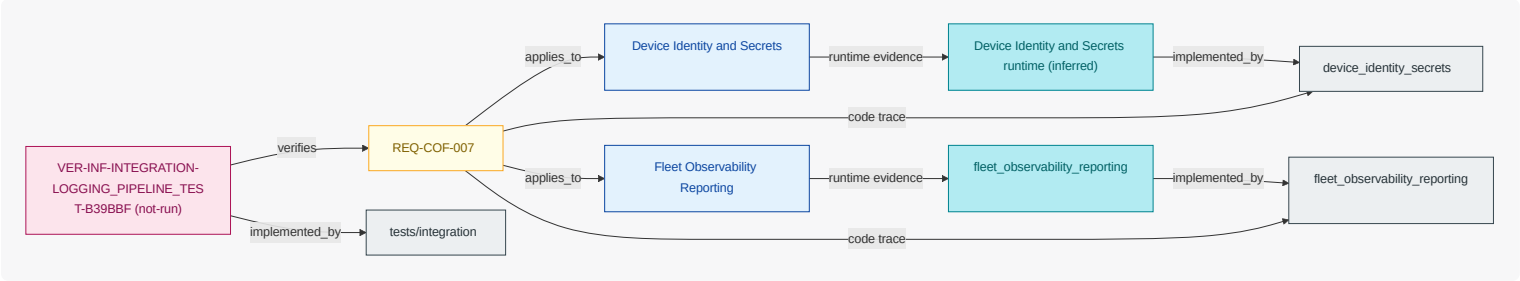
10.1.7. REQ-COF-007

Where `audit logging is enabled`, when `ota status reported event is received`, `the coffee fleet system` shall persist redacted `audit log record` with `machine identity` plus `campaign id`.

> **NOTE** Audit/compliance logging `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



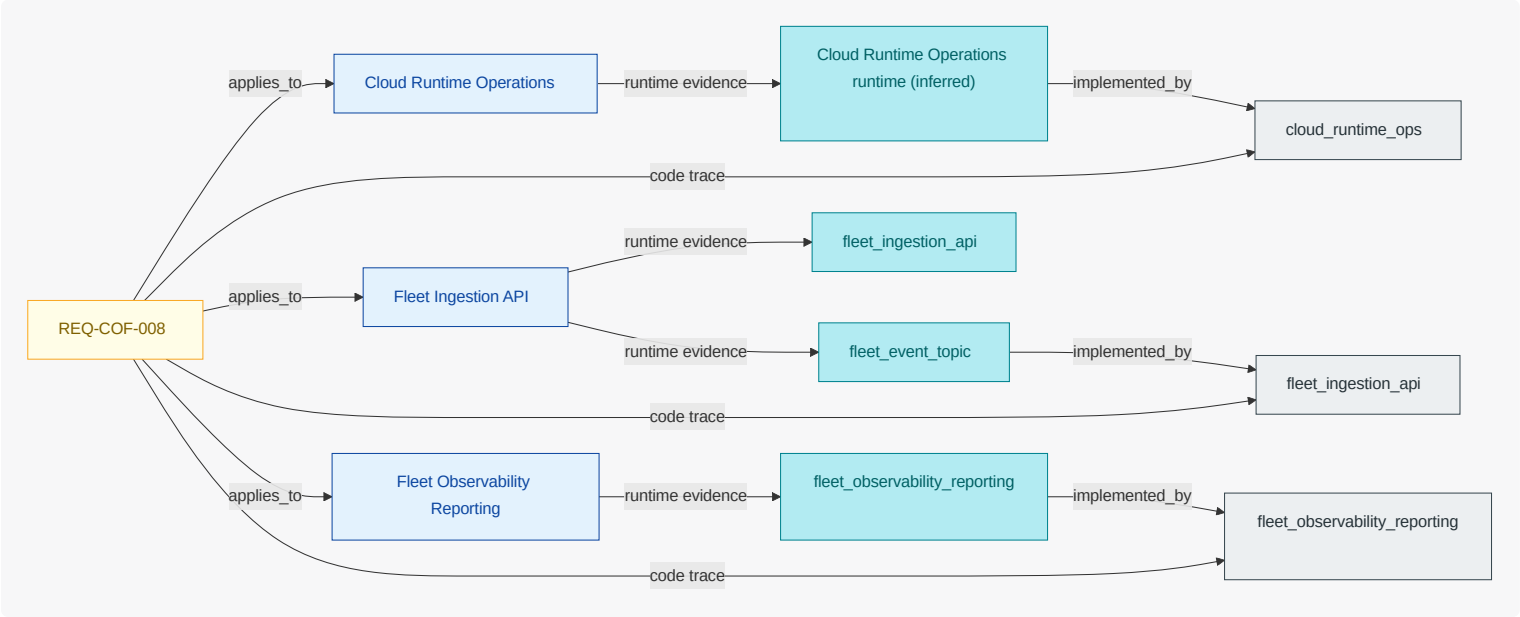
10.1.8. REQ-COF-008

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall throttle duplicate `telemetry`.

> **NOTE** Abuse control and security response `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



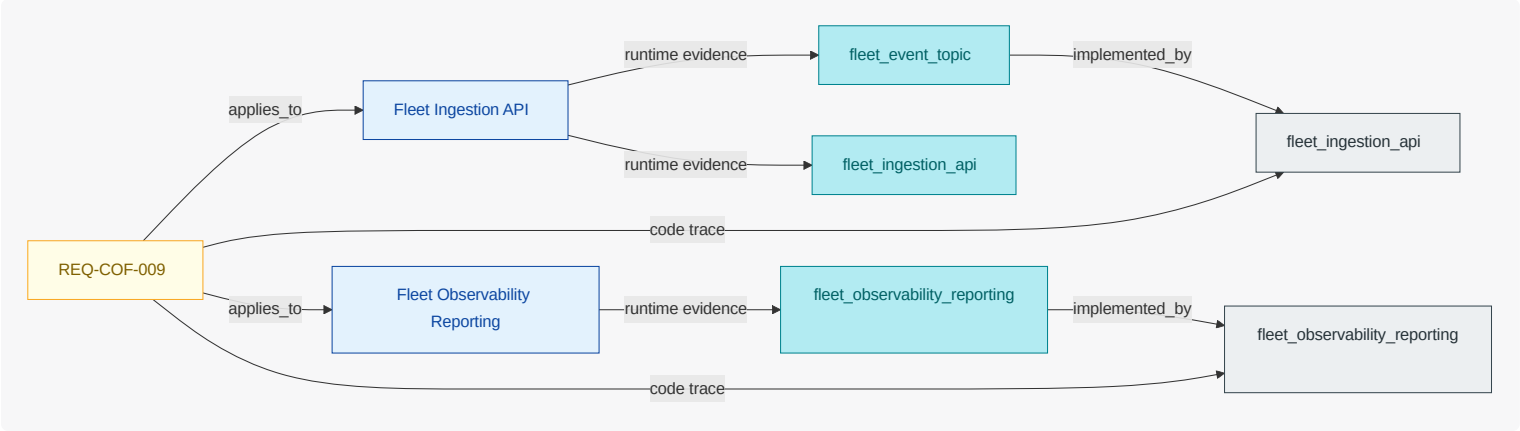
10.1.9. REQ-COF-009

While `coffee machine fleet operation is active`, when `telemetry ingested event is received`, `the coffee fleet system` shall persist `logging records` for `fleet operator`.

> **NOTE** Operational logging persistence `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



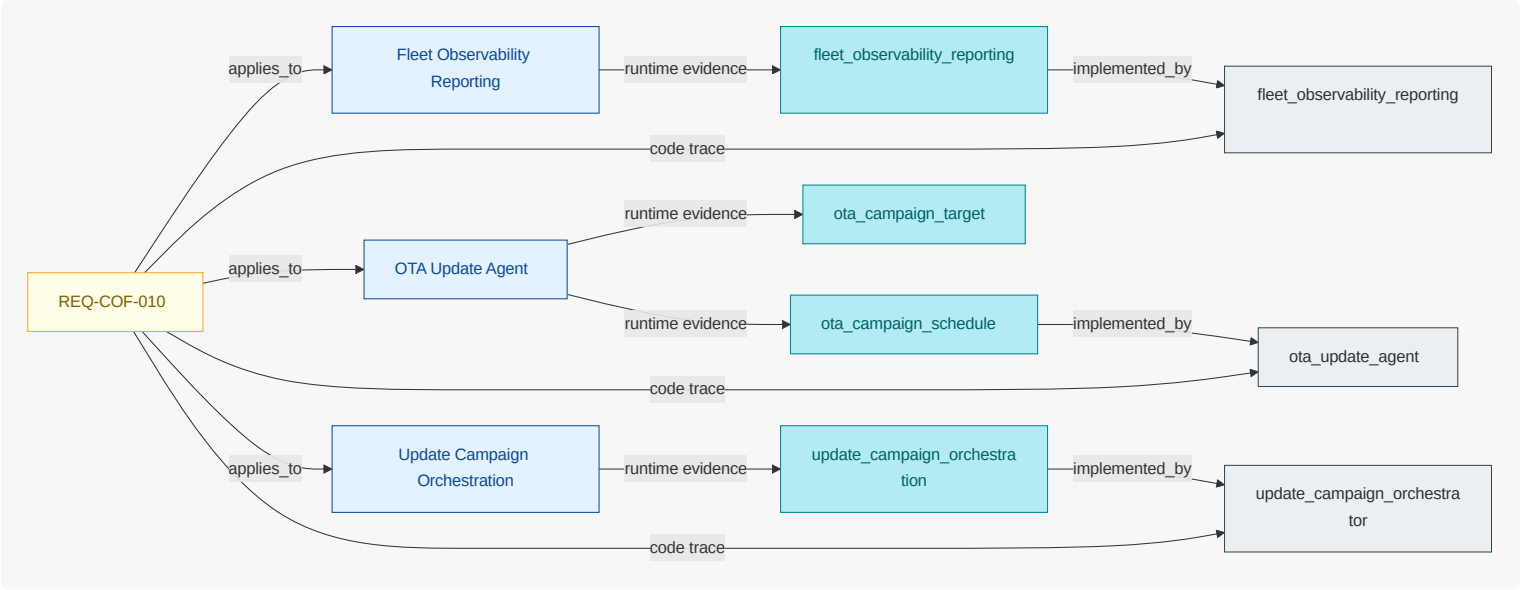
10.1.10. REQ-COF-010

If `firmware validation failed event is received`, then `the coffee fleet system` shall report `ota status` to `update campaign orchestration`.

> **NOTE** OTA failure status signaling `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



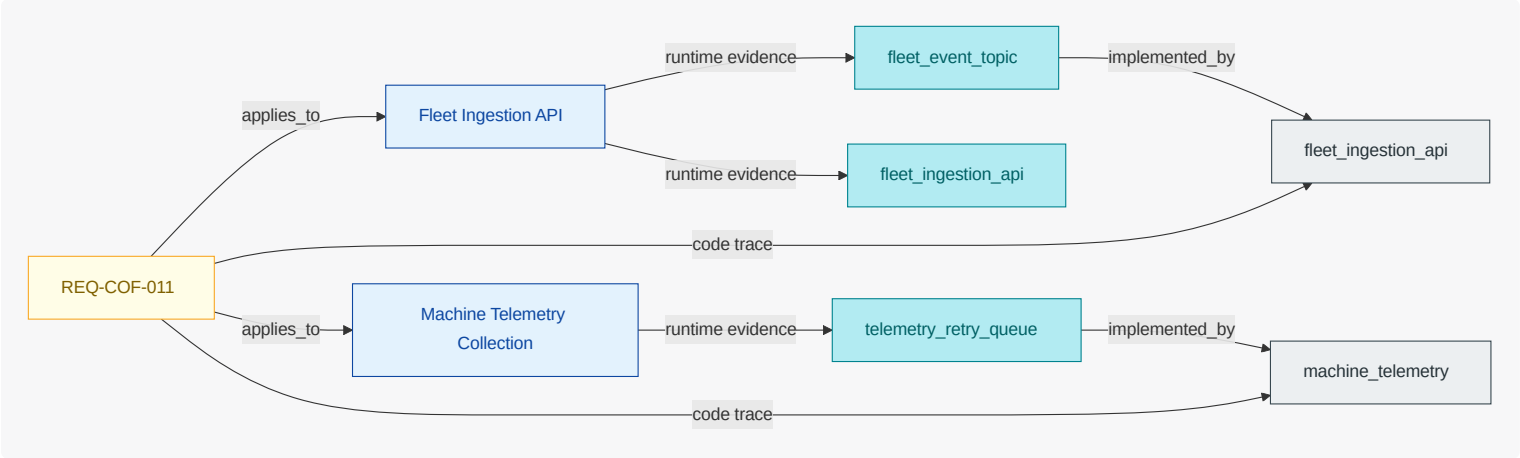
10.1.11. REQ-COF-011

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall flush queued `telemetry` when `normal mode is active`.

> **NOTE** Queued `telemetry` drain behavior after endpoint recovery.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



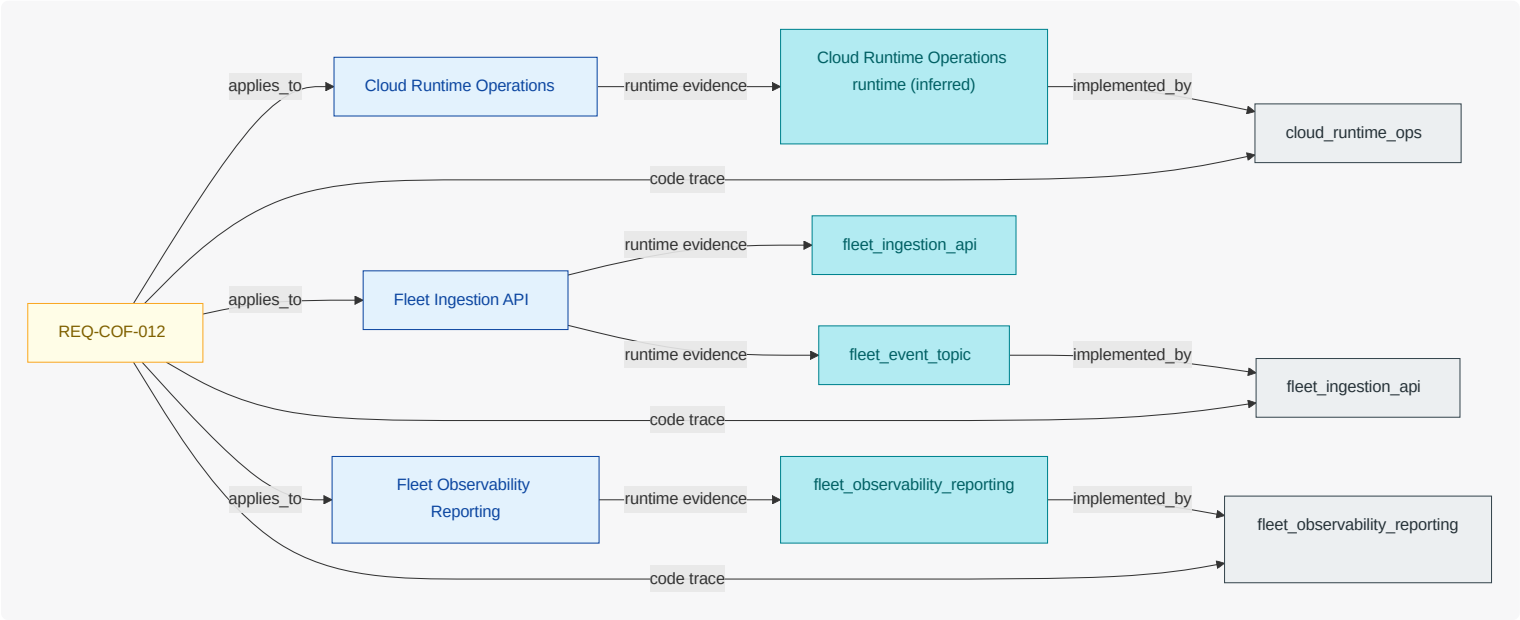
10.1.12. REQ-COF-012

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall notify `security analyst`.

> **NOTE** Security notification `requirement` for `replay abuse` events.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



10.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

10.2.1. Ota Rollout Flow

Field	Value
ID	`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`
Kind	e2e

Field	Value
Status	partial
Verifies Requirements	`REQ-COF-003`, `REQ-COF-005`
Test Code Elements	tests/e2e/ota_rollout_flow.yaml
Derived Functional Owners	`FU-FLEET-OBSERVABILITY-REPORTING`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	not-run:1, partial:1
Evidence	test-results/ota-rollback-report.md, tests/e2e/ota_rollout_flow.yaml
Notes	runs fleet `OTA rollout` scenario including scheduling, deployment, and `telemetry` confirmation

10.2.2. Logging Pipeline Test

Field	Value
ID	`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`
Kind	integration
Status	not-run
Verifies Requirements	`REQ-COF-002`, `REQ-COF-007`
Test Code Elements	tests/integration/logging_pipeline_test.ts
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-FLEET-INGESTION-API`, `FU-FLEET-OBSERVABILITY-REPORTING`
Result Summary	not-run:1, pass:1
Evidence	test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts
Notes	validates `telemetry` logging pipeline persistence and operational reporting signals

10.2.3. Telemetry Ingest Contract Test

Field	Value
ID	`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`
Kind	test
Status	pass
Verifies Requirements	`REQ-COF-001`
Test Code Elements	tests/contract/telemetry_ingest_contract_test.go

Field	Value
Derived Functional Owners	`FU-FLEET-INGESTION-API`, `FU-MACHINE-TELEMETRY-COLLECTION`
Result Summary	pass:1
Evidence	test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go
Notes	validates `telemetry` ingest contract fields and accepted request semantics

10.2.4. Ota Campaign Test

Field	Value
ID	`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`
Kind	unit
Status	pass
Verifies Requirements	`REQ-COF-003`, `REQ-COF-004`
Test Code Elements	tests/unit/ota_campaign_test.go
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	pass:2
Evidence	test-results/unit-ota.xml, tests/unit/ota_campaign_test.go
Notes	verifies OTA campaign `signature valid`ation and rollout trigger behavior

11. Reference Index

11.1. Authored References

REQ-COF-001

While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api.

Key	Value
Kind	Requirement
Name	REQ-COF-001
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Telemetry Ingest Contract Test`

REQ-COF-002

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-002
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Logging Pipeline Test`

REQ-COF-003

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort.

Key	Value
Kind	Requirement
Name	REQ-COF-003
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Ota Rollout Flow`, `Ota Campaign Test`

REQ-COF-004

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid.

Key	Value
Kind	Requirement
Name	REQ-COF-004

Key	Value
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Ota Campaign Test`

REQ-COF-005

If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply.

Key	Value
Kind	Requirement
Name	REQ-COF-005
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Ota Rollout Flow`

REQ-COF-006

If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry.

Key	Value
Kind	Requirement
Name	REQ-COF-006
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-007

Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record with machine identity plus campaign id.

Key	Value
Kind	Requirement
Name	REQ-COF-007
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`, `Logging Pipeline Test`

REQ-COF-008

If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry.

Key	Value
Kind	Requirement
Name	REQ-COF-008
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-009

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist logging records for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-009
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-010

If firmware validation failed event is received, then the coffee fleet system shall report ota status to update campaign orchestration.

Key	Value
Kind	Requirement
Name	REQ-COF-010
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-011

If cloud ingest endpoint is unavailable, then the coffee fleet system shall flush queued telemetry when normal mode is active.

Key	Value
Kind	Requirement
Name	REQ-COF-011
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-012

If telemetry replay abuse is detected, then the coffee fleet system shall notify security analyst.

Key	Value
Kind	Requirement
Name	REQ-COF-012
Mentioned In	`Architecture Intent View`, `Deployment View`

11.2. Catalog References

ACT-BARISTA

Operates coffee machines and observes local machine health.

Key	Value
Kind	Actor
Name	barista
Mentioned In	`Communication View`

ACT-FLEET-OPERATOR

Owns OTA rollout campaigns and fleet-level reliability.

Key	Value
Kind	Actor
Name	fleet operator
Mentioned In	`Communication View`, `REQ-COF-002`, `REQ-COF-009`

ACT-PLATFORM-OPERATOR

Owns runtime deployment and operational controls.

Key	Value
Kind	Actor
Name	platform operator
Mentioned In	`Communication View`

ACT-SECURITY-ANALYST

Reviews identity, firmware integrity, and logging controls.

Key	Value
Kind	Actor
Name	security analyst
Mentioned In	`Communication View`, `REQ-COF-012`

AV-LOG-TAMPERING

Attempts to suppress or alter security/operational log records.

Key	Value
Kind	Attack Vector
Name	log tampering
Mentioned In	n/a

AV-MALICIOUS-FIRMWARE-BUNDLE

Tampered firmware payload attempts to execute on devices.

Key	Value
Kind	Attack Vector
Name	malicious firmware bundle
Mentioned In	n/a

AV-SPOOFED-DEVICE-IDENTITY

Unauthorized actor impersonates a machine identity.

Key	Value
Kind	Attack Vector
Name	spoofed device identity
Mentioned In	n/a

AV-TELEMETRY-REPLAY-ABUSE

Replayed telemetry floods ingestion and distorts reporting.

Key	Value
Kind	Attack Vector
Name	telemetry replay abuse
Mentioned In	n/a

COND-FIRMWARE-SIGNATURE-VALID

Condition indicating firmware signature verification succeeded.

Key	Value
Kind	Condition
Name	firmware signature is valid
Mentioned In	`Security View`, `REQ-COF-004`, `Ota Campaign Test`

DATA-AUDIT-LOG-RECORD

Immutable log record for security and compliance analysis.

Key	Value
Kind	Data Term

Key	Value
Name	audit log record
Mentioned In	`Architecture Intent View`, `Communication View`, `Traceability View`, `REQ-COF-007`, `REQ-COF-009`

DATA-BREW-TELEMETRY

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Data Term
Name	brew telemetry
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-011`, `Ota Rollout Flow`, `Logging Pipeline Test`, `Telemetry Ingest Contract Test`

DATA-CAMPAIGN-ID

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Data Term
Name	campaign id
Mentioned In	`REQ-COF-007`

DATA-FIRMWARE-BUNDLE

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Data Term
Name	signed firmware bundle
Mentioned In	`Architecture Intent View`, `Communication View`, `Security View`, `REQ-COF-003`

DATA-MACHINE-IDENTITY

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Data Term
Name	machine identity
Mentioned In	`Communication View`, `Traceability View`, `REQ-COF-007`

DATA-OTA-CAMPAIGN-POLICY

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Data Term
Name	ota campaign policy
Mentioned In	`Architecture Intent View`, `Communication View`

DATA-OTA-STATUS

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Data Term
Name	ota status
Mentioned In	`Architecture Intent View`, `Communication View`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`

EM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	n/a

EM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

EM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	authored mapping
Mentioned In	`Communication View`, `Security View`

EM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

EM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

EM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	n/a

EM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

EM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

EM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	`Traceability View`, `REQ-COF-002`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`

EM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

EM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

EM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	verification check
Mentioned In	n/a

EVT-BREW-CYCLE-COMPLETED

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Event
Name	brew cycle completed event is received
Mentioned In	`REQ-COF-001`

EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Event
Name	cloud ingest endpoint is unavailable
Mentioned In	`Communication View`, `REQ-COF-006`, `REQ-COF-011`

EVT-FIRMWARE-VALIDATION-FAILED

Event raised when signature/hash validation fails on firmware.

Key	Value
Kind	Event
Name	firmware validation failed event is received
Mentioned In	`REQ-COF-005`, `REQ-COF-010`

EVT-OTA-CAMPAIGN-SCHEDULED

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Event
Name	ota campaign scheduled event is received
Mentioned In	`Communication View`, `REQ-COF-003`, `REQ-COF-004`

EVT-OTA-STATUS-REPORTED

Event raised when device reports OTA apply status.

Key	Value
Kind	Event
Name	ota status reported event is received
Mentioned In	`Communication View`, `REQ-COF-007`

EVT-TELEMETRY-INGESTED

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Event
Name	telemetry ingested event is received
Mentioned In	`Architecture Intent View`, `Communication View`, `REQ-COF-002`, `REQ-COF-009`

EVT-TELEMETRY-REPLAY-ABUSE-DETECTED

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Event
Name	telemetry replay abuse is detected
Mentioned In	`Security View`, `REQ-COF-008`, `REQ-COF-012`

FEAT-AUDIT-LOGGING

Feature flag enabling audit-grade log persistence.

Key	Value
Kind	Feature
Name	audit logging is enabled
Mentioned In	`REQ-COF-007`

FEAT-OTA-CAMPAIGN

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Feature
Name	ota campaign is enabled

Key	Value
Mentioned In	`Architecture Intent View`, `Communication View`, `REQ-COF-003`, `REQ-COF-004`, `Ota Rollout Flow`

FG-CLOUD-CONTROL

Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

Key	Value
Kind	Functional Group
Name	cloud control
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `REQ-COF-003`

FG-MACHINE-EDGE

Capability area for device-local telemetry and OTA execution behavior.

Key	Value
Kind	Functional Group
Name	machine edge
Mentioned In	`Architecture Intent View`, `Communication View`

FG-PLATFORM-OPERATIONS

Capability area for runtime operations and device identity/secrets lifecycle.

Key	Value
Kind	Functional Group
Name	platform operations
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`

FU-CLOUD-RUNTIME-OPERATIONS

Operates cloud runtime deployment and scaling controls.

Key	Value
Kind	Functional Unit
Name	cloud runtime operations
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

FU-DEVICE-IDENTITY-SECRETS

Manages device identity material and secure configuration.

Key	Value
Kind	Functional Unit
Name	device identity and secrets
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Logging Pipeline Test`, `Ota Campaign Test`

FU-FLEET-INGESTION-API

Ingests telemetry and machine status from edge devices.

Key	Value
Kind	Functional Unit
Name	fleet ingestion api
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `REQ-COF-001`, `Logging Pipeline Test`, `Telemetry Ingest Contract Test`

FU-FLEET-OBSERVABILITY-REPORTING

Aggregates telemetry and update outcomes into operational and audit signals.

Key	Value
Kind	Functional Unit
Name	fleet observability reporting
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Ota Rollout Flow`, `Logging Pipeline Test`

FU-MACHINE-TELEMETRY-COLLECTION

Collects brew/device telemetry and publishes cloud-bound payloads.

Key	Value
Kind	Functional Unit
Name	machine telemetry collection
Mentioned In	`Architecture Intent View`, `Communication View`, `Telemetry Ingest Contract Test`

FU-OTA-UPDATE-AGENT

Downloads, validates, and applies signed firmware updates on machines.

Key	Value
Kind	Functional Unit

Key	Value
Name	ota update agent
Mentioned In	`Architecture Intent View`, `Communication View`, `Ota Rollout Flow`, `Ota Campaign Test`

FU-UPDATE-CAMPAIGN-ORCHESTRATION

Selects target machine groups and coordinates OTA rollout phases.

Key	Value
Kind	Functional Unit
Name	update campaign orchestration
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `REQ-COF-010`, `Ota Rollout Flow`, `Ota Campaign Test`

MODE-NORMAL

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Mode
Name	normal mode is active
Mentioned In	`REQ-COF-011`

REF-CLOUD-LOGGING-SERVICE

Managed log and metrics service used for fleet observability.

Key	Value
Kind	Referenced Element
Name	cloud logging service
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

REF-CLOUD-RUNTIME-SDK

SDK used by cloud units for runtime integrations.

Key	Value
Kind	Referenced Element
Name	cloud runtime sdk
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

REF-FIRMWARE-BUNDLE-STORE

Object store containing signed OTA firmware bundles.

Key	Value
Kind	Referenced Element
Name	firmware bundle store
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

REF-FIRMWARE-SIGNER-SERVICE

Service that signs firmware artifacts before OTA rollout.

Key	Value
Kind	Referenced Element
Name	firmware signer service
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

REF-IOT-INGEST-ENDPOINT

Cloud endpoint for receiving telemetry and status from devices.

Key	Value
Kind	Referenced Element
Name	iot ingest endpoint
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

REF-MQTT-DEVICE-SDK

SDK used by edge components for telemetry and command transport.

Key	Value
Kind	Referenced Element
Name	mqtt device sdk
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`

STATE-FLEET-OPERATION-ACTIVE

State where machines and central cloud are actively serving operations.

Key	Value
Kind	State

Key	Value
Name	coffee machine fleet operation is active
Mentioned In	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-009`

SYS-COFFEE-FLEET-SYSTEM

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture.

Key	Value
Kind	System
Name	coffee fleet system
Mentioned In	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-006`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-011`, `REQ-COF-012`

11.3. Inferred Runtime References

ota_campaign_schedule

schedules OTA campaign orchestration triggers for configured rollout windows

Key	Value
Kind	Runtime eventbridge_rule
Owner	FU-OTA-UPDATE-AGENT
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

ota_campaign_target

routes scheduled OTA campaign events to the orchestration runtime handler

Key	Value
Kind	Runtime eventbridge_target
Owner	FU-OTA-UPDATE-AGENT
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

fleet_ingestion_api

receives device telemetry, validates payload shape, and publishes normalized fleet events

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-INGESTION-API
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

fleet_observability_reporting

aggregates telemetry outcomes into fleet operations and security reporting signals

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

update_campaign_orchestration

coordinates OTA campaign rollout windows and device cohort targeting

Key	Value
Kind	Runtime lambda_function
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

telemetry_retry_queue

buffers telemetry during ingest outages for retry once normal mode resumes

Key	Value
Kind	Runtime queue
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

fleet_event_topic

broadcasts fleet-wide telemetry and OTA lifecycle events to subscribed processors

Key	Value
Kind	Runtime topic
Owner	FU-FLEET-INGESTION-API
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Communication View`, `Deployment View`, `Security View`

11.4. Inferred Code References

cloud_runtime_ops.go

wires cloud runtime handlers and integration entrypoints

Key	Value
Kind	Code source_file
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	cloud_runtime_ops.go
Mentioned In	`Communication View`, `Deployment View`

device_identity_secrets.ts

resolves device identity credentials and firmware verification secrets

Key	Value
Kind	Code source_file
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts
Mentioned In	`Communication View`, `Deployment View`, `Security View`

fleet_ingestion_api.go

ingests device telemetry and emits normalized fleet ingestion events

Key	Value
Kind	Code source_file
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go
Mentioned In	n/a

fleet_observability_reporting.ts

aggregates telemetry and OTA outcome signals for fleet observability reporting

Key	Value
Kind	Code source_file
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts
Mentioned In	n/a

machine_telemetry.rs

builds machine telemetry and status payloads at the edge

Key	Value
Kind	Code source_file
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs
Mentioned In	`Communication View`, `Deployment View`, `Security View`

ota_update_agent.rs

applies OTA campaign updates and reports rollout outcomes

Key	Value
Kind	Code source_file
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs
Mentioned In	`Communication View`, `Deployment View`, `Security View`

update_campaign_orchestrator.ts

plans OTA rollout cohorts and campaign lifecycle transitions

Key	Value
Kind	Code source_file
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts
Mentioned In	`Communication View`, `Deployment View`, `Security View`

CODE-HANDLEROLLBACK

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:10
Mentioned In	n/a

CODE-INGESTTELEMETRY

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:11.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:11
Mentioned In	n/a

CODE-NOTIFYREPLAYABUSE

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:15.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:15
Mentioned In	n/a

CODE-PERSISTAUDITLOG

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:10
Mentioned In	n/a

CODE-PERSISTTELEMETRYMETRIC

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:5
Mentioned In	n/a

CODE-PLANCAMPAIGN

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:5
Mentioned In	n/a

CODE-PUBLISHBREWTELEMETRY

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:10.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:10
Mentioned In	n/a

CODE-QUEUEFORRETRY

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:15.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:15
Mentioned In	n/a

CODE-RAISESECURITYNOTIFICATION

Inferred code symbol owned by FU-CLOUD-RUNTIME-OPERATIONS from cloud_runtime_ops.go:6.

Key	Value
Kind	Code symbol
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	cloud_runtime_ops.go:6
Mentioned In	n/a

CODE-REDACTIDENTITYFORAUDIT

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:10
Mentioned In	n/a

CODE-REJECTREPLAY

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:16.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:16
Mentioned In	n/a

CODE-ROLLBACKANDREPORT

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:14.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:14
Mentioned In	n/a

CODE-VALIDATEANDAPPLY

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:5.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:5
Mentioned In	n/a

CODE-VERIFYFIRMWARESIGNATURE

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:5
Mentioned In	n/a

11.5. Verification References

VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

Key	Value
Kind	e2e
Status	partial
Source	test-results/ota-rollback-report.md (+1)
Name	Ota Rollout Flow
Mentioned In	`Traceability View`, `Ota Rollout Flow`

VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF

validates telemetry logging pipeline persistence and operational reporting signals

Key	Value
Kind	integration
Status	not-run
Source	test-results/integration-logging.json (+1)

Key	Value
Name	Logging Pipeline Test
Mentioned In	`Traceability View`, `Logging Pipeline Test`

VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794

validates telemetry ingest contract fields and accepted request semantics

Key	Value
Kind	test
Status	pass
Source	test-results/contract-telemetry.json (+1)
Name	Telemetry Ingest Contract Test
Mentioned In	`Traceability View`, `Telemetry Ingest Contract Test`

VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8

verifies OTA campaign signature validation and rollout trigger behavior

Key	Value
Kind	unit
Status	pass
Source	test-results/unit-ota.xml (+1)
Name	Ota Campaign Test
Mentioned In	`Traceability View`, `Ota Campaign Test`